

DATABASE MANAGEMENT SYSTEMS

(ABHAY KUMAR 25BCE2537)

ASSIGNMENT – 2

1. CREATION OF DATABASE AND INSERTION OF VALUES

```
-- PRODUCT ORDER MANAGEMENT SYSTEM
-- Name: ABHAY KUMAR
-- Reg No: 25BCE2537

● CREATE DATABASE ProductOrderDB;
● USE ProductOrderDB;

-- 1. CUSTOMERS TABLE

● CREATE TABLE CUSTOMERS (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(50),
    Phone_Number VARCHAR(15),
    Email_Address VARCHAR(60),
    City VARCHAR(30)
);

● INSERT INTO CUSTOMERS VALUES
(101, 'Rahul Sharma', '9876543210', 'rahul.sharma@gmail.com', 'Chennai'),
(102, 'Priya Nair', '9123456780', 'priya.nair@gmail.com', 'Bengaluru'),
(103, 'Arjun Mehta', '9988776655', 'arjun.mehta@gmail.com', 'Mumbai');

-- 2. PRODUCTS TABLE

● CREATE TABLE PRODUCTS (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Category VARCHAR(30),
    Unit_Price DECIMAL(10,2),
    Stock_Quantity INT
);
```

Limit to 1000 rows

Jump to

```
34 INSERT INTO PRODUCTS VALUES
35 (201, 'Wireless Mouse', 'Electronics', 699.00, 50),
36 (202, 'Mechanical Keyboard', 'Electronics', 2499.00, 30),
37 (203, 'USB-C Charger', 'Accessories', 999.00, 75);
38
39 3. ORDERS TABLE
40
41 CREATE TABLE ORDERS (
42     Order_ID INT PRIMARY KEY,
43     Order_Date DATE,
44     Order_Status VARCHAR(20),
45     Delivery_Address VARCHAR(100),
46     Customer_ID INT,
47     FOREIGN KEY (Customer_ID) REFERENCES CUSTOMERS(Customer_ID)
48 );
49
50 INSERT INTO ORDERS VALUES
51 (301, '2026-07-20', 'Delivered', '12 Anna Nagar, Chennai', 101),
52 (302, '2026-07-22', 'Shipped', '45 Indiranagar, Bengaluru', 102),
53 (303, '2026-07-24', 'Processing', '8 Andheri West, Mumbai', 103);
54
55 -- 4. ORDER_ITEMS TABLE
56 CREATE TABLE ORDER_ITEMS (
57     Item_ID INT PRIMARY KEY,
58     Order_ID INT,
59     Product_ID INT,
60     Quantity INT,
61     Price_Each DECIMAL(10,2),
62     FOREIGN KEY (Order_ID) REFERENCES ORDERS(Order_ID),
63     FOREIGN KEY (Product_ID) REFERENCES PRODUCTS(Product_ID)
64 );
65 INSERT INTO ORDER_ITEMS VALUES
66 (401, 301, 201, 2, 699.00),
```

Automatic context help is disabled. Use the toolbar
caret position or to toggle :

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 1	18:41:28	CREATE DATABASE ProductOrderDB	1 row(s) affected
✓ 2	18:41:28	USE ProductOrderDB	0 row(s) affected
✓ 3	18:41:28	CREATE TABLE CUSTOMERS (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), ...	0 row(s) affected
✓ 4	18:41:28	INSERT INTO CUSTOMERS VALUES (101, 'Rahul Shama', '9876543210', 'rahul.shama@gmail.com', 'Chen...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 5	18:41:28	CREATE TABLE PRODUCTS (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(50), Cate...	0 row(s) affected
✓ 6	18:41:28	INSERT INTO PRODUCTS VALUES (201, 'Wireless Mouse', 'Electronics', 699.00, 50), (202, 'Mechanical Key...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Limit to 1000 rows

Jump to

```
39 3. ORDERS TABLE
40
41 CREATE TABLE ORDERS (
42     Order_ID INT PRIMARY KEY,
43     Order_Date DATE,
44     Order_Status VARCHAR(20),
45     Delivery_Address VARCHAR(100),
46     Customer_ID INT,
47     FOREIGN KEY (Customer_ID) REFERENCES CUSTOMERS(Customer_ID)
48 );
49
50 INSERT INTO ORDERS VALUES
51 (301, '2026-07-20', 'Delivered', '12 Anna Nagar, Chennai', 101),
52 (302, '2026-07-22', 'Shipped', '45 Indiranagar, Bengaluru', 102),
53 (303, '2026-07-24', 'Processing', '8 Andheri West, Mumbai', 103);
54
55 -- 4. ORDER_ITEMS TABLE
56 CREATE TABLE ORDER_ITEMS (
57     Item_ID INT PRIMARY KEY,
58     Order_ID INT,
59     Product_ID INT,
60     Quantity INT,
61     Price_Each DECIMAL(10,2),
62     FOREIGN KEY (Order_ID) REFERENCES ORDERS(Order_ID),
63     FOREIGN KEY (Product_ID) REFERENCES PRODUCTS(Product_ID)
64 );
65 INSERT INTO ORDER_ITEMS VALUES
66 (401, 301, 201, 2, 699.00),
67 (402, 301, 203, 1, 999.00),
68 (403, 302, 202, 1, 2499.00),
69 (404, 303, 201, 1, 699.00);
70
71
```

Automatic context help is disabled. Use the t
caret position or to to

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 5	18:41:28	CREATE TABLE PRODUCTS (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(50), Cate...	0 row(s) affected
✓ 6	18:41:28	INSERT INTO PRODUCTS VALUES (201, 'Wireless Mouse', 'Electronics', 699.00, 50), (202, 'Mechanical Key...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	18:41:28	CREATE TABLE ORDERS (Order_ID INT PRIMARY KEY, Order_Date DATE, Order_Status VARCHA...	0 row(s) affected
✓ 8	18:41:28	INSERT INTO ORDERS VALUES (301, '2026-07-20', 'Delivered', '12 Anna Nagar, Chennai', 101), (302, '2026-...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 9	18:41:28	CREATE TABLE ORDER_ITEMS (Item_ID INT PRIMARY KEY, Order_ID INT, Product_ID INT, Q...	0 row(s) affected
✓ 10	18:41:28	INSERT INTO ORDER_ITEMS VALUES (401, 301, 201, 2, 699.00), (402, 301, 203, 1, 999.00), (403, 302, 20...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

2. 17 QUERIES PERFORMED ON DATABASE

A.

```
1 --
2 -- PRODUCT ORDER MANAGEMENT SYSTEM
3 -- Name : ABHAY KUMAR
4 -- Reg No: 25BCE2537
5 -- Queries 1 to 17
6
7 • USE ProductOrderDB;
8 -- QUERY 1: Display all customers
9
10 • SELECT * FROM CUSTOMERS;
11
```

Customer_ID	Customer_Name	Phone_Number	Email_Address	City
101	Rahul Sharma	9876543210	rahul.sharma@gmail.com	Chennai
102	Priya Nair	9123456780	priya.nair@gmail.com	Bengaluru
103	Arjun Mehta	9988776655	arjun.mehta@gmail.com	Mumbai
NULL	NULL	NULL	NULL	NULL

B.

```
1
2 -- QUERY 2: Display all products
3 • SELECT * FROM PRODUCTS;
4
```

Product_ID	Product_Name	Category	Unit_Price	Stock_Quantity
201	Wireless Mouse	Electronics	699.00	50
202	Mechanical Keyboard	Electronics	2499.00	30
203	USB-C Charger	Accessories	999.00	75
NULL	NULL	NULL	NULL	NULL

C.

```
1
2
3 -- QUERY 3: Display all orders
4 • SELECT * FROM ORDERS;
5
```

Order_ID	Order_Date	Order_Status	Delivery_Address	Customer_ID
301	2026-07-20	Delivered	12 Anna Nagar, Chennai	101
302	2026-07-22	Shipped	45 Indiranagar, Bengaluru	102
303	2026-07-24	Processing	8 Andheri West, Mumbai	103
NULL	NULL	NULL	NULL	NULL

D.

```
17
18 -- QUERY 4: Display all order items
19 • SELECT * FROM ORDER_ITEMS;
20
21
```

	Item_ID	Order_ID	Product_ID	Quantity	Price_Each
▶	401	301	201	2	699.00
	402	301	203	1	999.00
	403	302	202	1	2499.00
	404	303	201	1	699.00
*	NULL	NULL	NULL	NULL	NULL

E AND F

```
-- QUERY 5: Display customer names and cities
-- =====
SELECT Customer_Name, City
FROM CUSTOMERS;

-- =====
-- QUERY 6: Products with price greater than 1000
-- =====
SELECT Product_Name, Unit_Price
FROM PRODUCTS
WHERE Unit_Price > 1000;
```

	Customer_Name	City
▶	Rahul Sharma	Chennai
	Priya Nair	Bengaluru
	Arjun Mehta	Mumbai

	Product_Name	Unit_Price
▶	Mechanical Keyboard	2499.00

G AND H

```
-- QUERY 7: Orders with status 'Delivered'
-- =====
• SELECT Order_ID, Order_Status
  FROM ORDERS
 WHERE Order_Status = 'Delivered';
|
-- QUERY 8: Products sorted by price (highest first)
• SELECT Product_Name, Unit_Price
  FROM PRODUCTS
 ORDER BY Unit_Price DESC;
```

Result Grid | Filter Rows:

	Order_ID	Order_Status
▶	301	Delivered
*	NULL	NULL

Result Grid | Filter Rows:

	Product_Name	Unit_Price
▶	Mechanical Keyboard	2499.00
	USB-C Charger	999.00
	Wireless Mouse	699.00

I AND J

```
-- QUERY 9: Total number of customers
-- =====
• SELECT COUNT(*) AS Total_Customers
  FROM CUSTOMERS;

-- =====
-- QUERY 10: Total stock quantity
-- =====
• SELECT SUM(Stock_Quantity) AS Total_Stock
  FROM PRODUCTS;
```

Total_Customers
4

Total_Stock
155

K AND L

```
-- QUERY 11: Average product price
-- =====

SELECT AVG(Unit_Price) AS Average_Price
FROM PRODUCTS;

-- =====

-- QUERY 12: Highest priced product
-- =====

SELECT Product_Name, Unit_Price
FROM PRODUCTS
WHERE Unit_Price = (SELECT MAX(Unit_Price) FROM PRODUCTS);
```

Average_Price
1399.000000

Product_Name	Unit_Price
Mechanical Keyboard	2499.00

M AND N

```
80 -- QUERY 13: Orders with customer names
81 • SELECT O.Order_ID,
82         O.Order_Date,
83         C.Customer_Name
84 FROM ORDERS O
85 JOIN CUSTOMERS C
86 ON O.Customer_ID = C.Customer_ID;
87
88 -- =====
```

Order_ID	Order_Date	Customer_Name
301	2026-07-20	Rahul Sharma
302	2026-07-22	Priya Nair
303	2026-07-24	Arjun Mehta

```
-- QUERY 14: Ordered products with quantity
-- =====
SELECT OI.Item_ID,
       P.Product_Name,
       OI.Quantity
FROM ORDER_ITEMS OI
JOIN PRODUCTS P
ON OI.Product_ID = P.Product_ID;
```

O AND P

```
98  -- =====
99  -- QUERY 15: Line total for each order item
100 -- =====
101 • SELECT Item_ID,
102        Quantity,
103        Price_Each,
104        Quantity * Price_Each AS Line_Total
105 FROM ORDER_ITEMS;
106
```

Result Grid					Filter Rows:		Export:	Wrap Cell
	Item_ID	Quantity	Price_Each	Line_Total				
▶	401	2	699.00	1398.00				
	402	1	999.00	999.00				
	403	1	2499.00	2499.00				
	404	1	699.00	699.00				

```
107 -- =====
108 -- QUERY 16: Total amount for each order
109 -- =====
110 • SELECT Order_ID,
111        SUM(Quantity * Price_Each) AS Order_Total
112 FROM ORDER_ITEMS
113 GROUP BY Order_ID;
114
115 -- =====
```

Result Grid			Filter Rows:		Export:	Wrap Cell Content
	Order_ID	Order_Total				
▶	301	2397.00				
	302	2499.00				
	303	699.00				

Q.

```
117  -- =====
118  ●  SELECT O.Order_ID,
119         C.Customer_Name,
120         P.Product_Name,
121         OI.Quantity,
122         OI.Price_Each,
123         O.Order_Status
124  FROM ORDERS O
125  JOIN CUSTOMERS C
126  ON O.Customer_ID = C.Customer_ID
127  JOIN ORDER_ITEMS OI
128  ON O.Order_ID = OI.Order_ID
129  JOIN PRODUCTS P
130  ON OI.Product_ID = P.Product_ID;
131
```

Order_ID	Customer_Name	Product_Name	Quantity	Price_Each	Order_Status
301	Rahul Sharma	Wireless Mouse	2	699.00	Delivered
301	Rahul Sharma	USB-C Charger	1	999.00	Delivered
302	Priya Nair	Mechanical Keyboard	1	2499.00	Shipped
303	Arjun Mehta	Wireless Mouse	1	699.00	Processing

INSERTION QUERIES

```
1  ●  USE ProductOrderDB;
2  INSERT INTO CUSTOMERS VALUES
3  (106, 'Amit Kumar', '9898989898', 'amit.kumar@gmail.com', 'Pune');
4  ●  INSERT INTO PRODUCTS VALUES
5  (206, 'Wireless Mouse', 'Electronics', 699.00, 50);
6  ●  INSERT INTO ORDERS VALUES
7  (306, '2026-07-20', 'Delivered', '12 Anna Nagar, Chennai', 101);
8  ●  INSERT INTO ORDER_ITEMS VALUES
9  (406, 301, 201, 2, 699.00);
10 ●  INSERT INTO PRODUCTS VALUES
11 (206, 'Mechanical Keyboard', 'Electronics', 2499.00, 30);
```

Customer_ID	Customer_Name	Phone_Number	Email_Address	City
101	Rahul Sharma	9876543210	rahul.sharma@gmail.com	Chennai
102	Priya Nair	9123456780	priya.nair@gmail.com	Bengaluru
103	Arjun Mehta	9988776655	arjun.mehta@gmail.com	Mumbai
106	Amit Kumar	9898989898	amit.kumar@gmail.com	Pune
NULL	NULL	NULL	NULL	NULL

CUSTOMERS 5 × PRODUCTS 6 ORDERS 7 ORDER_ITEMS 8 Apply Revert

Product_ID	Product_Name	Category	Unit_Price	Stock_Quantity
201	Wireless Mouse	Electronics	699.00	50
202	Mechanical Keyboard	Electronics	2499.00	30
203	USB-C Charger	Accessories	999.00	75
NULL	NULL	NULL	NULL	NULL

CUSTOMERS 5 PRODUCTS 6 × ORDERS 7 ORDER_ITEMS 8 Apply

Order_ID	Order_Date	Order_Status	Delivery_Address	Customer_ID
301	2026-07-20	Delivered	12 Anna Nagar, Chennai	101
302	2026-07-22	Shipped	45 Indiranagar, Bengaluru	102
303	2026-07-24	Processing	8 Andheri West, Mumbai	103
NULL	NULL	NULL	NULL	NULL

Item_ID	Order_ID	Product_ID	Quantity	Price_Each
401	301	201	2	699.00
402	301	203	1	999.00
403	302	202	1	2499.00
404	303	201	1	699.00
NULL	NULL	NULL	NULL	NULL

ASSIGNMENT QUERIES: -

1.

```
USE ProductOrderDB;
-- ASSIGNMENT 1
-- Retrieve all customer details
SELECT * FROM CUSTOMERS;
```

Customer_ID	Customer_Name	Phone_Number	Email_Address	City
101	Rahul Sharma	9876543210	rahul.sharma@gmail.com	Chennai
102	Priya Nair	9123456780	priya.nair@gmail.com	Bengaluru
103	Arjun Mehta	9988776655	arjun.mehta@gmail.com	Mumbai
106	Amit Kumar	9898989898	amit.kumar@gmail.com	Pune
NULL	NULL	NULL	NULL	NULL

2.

```
-- ASSIGNMENT 2
-- Display customer name along with order details
SELECT C.Customer_Name,
       O.Order_ID,
       O.Order_Date,
       O.Order_Status
FROM CUSTOMERS C
JOIN ORDERS O
ON C.Customer_ID = O.Customer_ID;
```

	Customer_Name	Order_ID	Order_Date	Order_Status
▶	Rahul Sharma	301	2026-07-20	Delivered
	Priya Nair	302	2026-07-22	Shipped
	Arjun Mehta	303	2026-07-24	Processing

3.

```
--
12 -- QUERY 3: Display records with a condition (City = 'Chennai')
13 • SELECT * FROM CUSTOMERS
14 WHERE City = 'Chennai';
15
16 -- QUERY 4: Display records for a particular year (2026)
17 • SELECT * FROM ORDERS
18 WHERE YEAR(Order_Date) = 2026;
19
20 -- QUERY 5: Display only the first three records from any table (
```

Result Grid					
Filter Rows:					
Edit:    Export/Import 					
	Customer_ID	Customer_Name	Phone_Number	Email_Address	City
▶	101	Rahul Sharma	9876543210	rahul.sharma@gmail.com	Chennai

4.

```
25 -- ASSIGNMENT 4
26 -- Display customer name, product purchased and quantity
27 • SELECT C.Customer_Name,
28       P.Product_Name,
29       OI.Quantity
30 FROM CUSTOMERS C
31 JOIN ORDERS O
32 ON C.Customer_ID = O.Customer_ID
33 JOIN ORDER_ITEMS OI
34 ON O.Order_ID = OI.Order_ID
35 JOIN PRODUCTS P
36 ON OI.Product_ID = P.Product_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Customer_Name	Product_Name	Quantity
Rahul Sharma	Wireless Mouse	2
Rahul Sharma	USB-C Charger	1
Priya Nair	Mechanical Keyboard	1
Arjun Mehta	Wireless Mouse	1

5.

```
15
16 -- QUERY 4: Display records for a particular year (2026)
17 • SELECT * FROM ORDERS
18 WHERE YEAR(Order_Date) = 2026;
19
20 -- QUERY 5: Display only the first three records from any table
```

Result Grid | Filter Rows: | Edit: | Export/Import

Order_ID	Order_Date	Order_Status	Delivery_Address	Customer_ID
301	2026-07-20	Delivered	12 Anna Nagar, Chennai	101
302	2026-07-22	Shipped	45 Indiranagar, Bengaluru	102
303	2026-07-24	Processing	8 Andheri West, Mumbai	103
NULL	NULL	NULL	NULL	NULL

6.

```
44  -- ASSIGNMENT 6
45  -- Find products belonging to the Electronics category
46  ●  SELECT Product_ID,
47         Product_Name,
48         Category
49  FROM PRODUCTS
50  WHERE Category = 'Electronics';
```

Result Grid			
Filter Rows:			
Edit:   			
	Product_ID	Product_Name	Category
▶	201	Wireless Mouse	Electronics
	202	Mechanical Keyboard	Electronics
✱	NULL	NULL	NULL